

```

mysql> // =====
-> // TASK 1 : DOCUMENT MODELLING
-> // =====
->
-> db.book_metadata.insertOne({
->   title: "The Silent Algorithm",
->   author: "R. Menon",
->   genre: "Technology",
->   publishedYear: 2022,
->   formats: ["Hardcover", "Digital", "Audiobook"],
->   specs: {
->     pages: 340,
->     fileSize_MB: 4.2,
->     drm: true,
->     narrator: "K. Rao",
->     runtime_min: 610
->   },
->   avgRating: 4.6,
->   reviews: [
->     {
->       member_id: "M1001",
->       rating: 5,
->       comment: "Excellent deep dive into digital systems design.",
->       date: ISODate("2023-01-15")
->     },
->     {
->       member_id: "M1042",
->       rating: 4,
->       comment: "Good but the Digital edition had formatting issues.",
->       date: ISODate("2023-02-20")
->     }
->   ]
-> });
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ =====
// TASK 1 : DOCUMENT M' at line 1
mysql>
mysql> db.book_metadata.insertMany([
-> {
->   title: "Ocean of Code",
->   author: "A. Fernandes",

```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ =====
// TASK 1 : DOCUMENT M' at line 1
mysql>
mysql> db.book_metadata.insertMany([
  -> {
  ->   title: "Ocean of Code",
  ->   author: "A. Fernandes",
  ->   genre: "Technology",
  ->   publishedYear: 2021,
  ->   formats: ["Digital", "Paperback"],
  ->   specs: {
  ->     pages: 210,
  ->     fileSize_MB: 2.1,
  ->     drm: false
  ->   },
  ->   avgRating: 4.2,
  ->   reviews: [
  ->     {
  ->       member_id: "M2001",
  ->       rating: 4,
  ->       comment: "Solid Digital read, clean layout.",
  ->       date: ISODate("2022-05-10")
  ->     },
  ->     {
  ->       member_id: "M2002",
  ->       rating: 5,
  ->       comment: "Best programming book this year.",
  ->       date: ISODate("2022-06-01")
  ->     }
  ->   ]
  -> },
  -> {
  ->   title: "Whispers of Yesterday",
  ->   author: "S. Kapoor",
  ->   genre: "Fiction",
  ->   publishedYear: 2019,
  ->   formats: ["Hardcover", "Paperback"],
  ->   specs: {
  ->     pages: 410,
  ->     binding: "Sewn"
```

```
->     binding: "Sewn"
->   },
->   avgRating: 3.8,
->   reviews: [
->     {
->       member_id: "M3001",
->       rating: 3,
->       comment: "Slow start but strong ending.",
->       date: ISODate("2020-01-15")
->     },
->     {
->       member_id: "M3002",
->       rating: 4,
->       comment: "Beautifully written prose.",
->       date: ISODate("2020-02-11")
->     }
->   ]
-> },
-> {
->   title: "Quantum Horizons",
->   author: "R. Menon",
->   genre: "Science",
->   publishedYear: 2023,
->   formats: ["Digital", "Audiobook"],
->   specs: {
->     fileSize_MB: 5.6,
->     drm: true,
->     narrator: "P. Iyer",
->     runtime_min: 540
->   },
->   avgRating: 4.9,
->   reviews: [
->     {
->       member_id: "M4001",
->       rating: 5,
->       comment: "The Digital format made annotations so easy.",
->       date: ISODate("2023-08-01")
->     },
->     {
->       member_id: "M4002",
->       rating: 5,
```

```

->         rating: 5,
->         comment: "Mind-bending and well narrated.",
->         date: ISODate("2023-09-14")
->     }
-> ]
-> }
-> ]});
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'd
b.book_metadata.insertMany([
{
    title: "Ocean of Code",
    author: "A. Fern' at line 1
mysql>
mysql> // =====
-> // TASK 2 : ADVANCED QUERYING
-> // =====
->
-> // Books rated above 4
-> db.book_metadata.find(
->     { avgRating: { $gt: 4 } },
->     { title: 1, avgRating: 1, _id: 0 }
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ =====
// TASK 2 : ADVANCED Q' at line 1
mysql>
mysql> // Books in Digital or Audiobook formats
-> db.book_metadata.find(
->     { formats: { $in: ["Digital", "Audiobook"] } },
->     { title: 1, formats: 1, _id: 0 }
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Books in Digital or Audiobook formats
db.book_metadata.find(
    { formats: {' at line 1
mysql>
mysql> // Combined query
-> db.book_metadata.find({
->     avgRating: { $gt: 4 },
->     formats: { $in: ["Digital"] }
-> });

```

```

-> });
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Combined query
db.book_metadata.find(
  avgRating: { $gt: 4 },
  formats' at line 1
mysql>
mysql> // Regex query on formats
-> db.book_metadata.find(
->   { formats: { $regex: "Digital", $options: "i" } },
->   { title: 1, formats: 1 }
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Regex query on formats
db.book_metadata.find(
  { formats: { $regex: "Digit' at line 1
mysql>
mysql> // Regex query on reviews
-> db.book_metadata.find(
->   { "reviews.comment": { $regex: "Digital", $options: "i" } },
->   { title: 1, "reviews.$": 1 }
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Regex query on reviews
db.book_metadata.find(
  { "reviews.comment": { $reg' at line 1
mysql>
mysql> // =====
-> // TASK 3 : AGGREGATION PIPELINE
-> // =====
->
-> db.book_metadata.aggregate([
->
->   { $match: { publishedYear: { $gt: 2020 } } },
->
->   { $unwind: "$reviews" },
->
->   {
->     $group: {
->       _id: "$title",
->       avgReviewRating: { $avg: "$reviews.rating" },

```

```

->         avgReviewRating: { $avg: "$reviews.rating" },
->         totalReviews: { $sum: 1 },
->         author: { $first: "$author" }
->     }
-> },
->
-> { $sort: { avgReviewRating: -1 } },
->
-> {
->     $project: {
->         _id: 0,
->         title: "$_id",
->         author: 1,
->         avgReviewRating: { $round: ["$avgReviewRating", 2] },
->         totalReviews: 1
->     }
-> }
->
-> ]});
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ =====
// TASK 3 : AGGREGATIO' at line 1
mysql>
mysql> // =====
-> // TASK 4 : INDEXING
-> // =====
->
-> // Create Text Index
-> db.book_metadata.createIndex({
->     "reviews.comment": "text"
-> });
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ =====
// TASK 4 : INDEXING
/' at line 1
mysql>
mysql> // Search using text index
-> db.book_metadata.find(
->     { $text: { $search: "Digital" } },
->     { title: 1, score: { $meta: "textScore" } }
-> ).sort({

```

```

->       score: { $meta: "textScore" }
-> });
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Search using text index
db.book_metadata.find(
  { $text: { $search: "Digit" } } at line 1
mysql>
mysql> // Verify index usage
-> db.book_metadata.find(
->   { $text: { $search: "Digital" } }
-> ).explain("executionStats");
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Verify index usage
db.book_metadata.find(
  { $text: { $search: "Digital" } }' at line 1
mysql>
mysql> // Regex scan comparison
-> db.book_metadata.find(
->   { "reviews.comment": { $regex: "Digital", $options: "i" } }
-> ).explain("executionStats");
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Regex scan comparison
db.book_metadata.find(
  { "reviews.comment": { $rege' at line 1
mysql>
mysql> // Multikey Index
-> db.book_metadata.createIndex({ formats: 1 });
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Multikey Index
db.book_metadata.createIndex({ formats: 1 })' at line 1
mysql>
mysql> // Rating Index
-> db.book_metadata.createIndex({ avgRating: -1 });
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
/ Rating Index
db.book_metadata.createIndex({ avgRating: -1 })' at line 1
mysql>
mysql> // Test rating index
-> db.book_metadata.find({
->   avgRating: { $gt: 4 }
-> }).explain("executionStats");|

```